

Software Test Plan (STP) - Event Management System

Project: Event Management System

Version: 1.0

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Status: Draft

1. Introduction

Purpose: This document defines the test plan for the Event Management System. It describes the scope and approach for testing the system to ensure that its major functions work as specified in the SRS.

Scope: Testing covers the major functionalities of the Event Management System, including user authentication and role management, event creation and management, event discovery and search, event registration, registration history, participation management, attendance tracking, event status and capacity management, and reporting.

References: The main references for testing are the Event Management System SRS , the project proposal, and the Requirements Traceability Matrix associated with the project.

Definitions: PESU-EMS refers to the PES University Event Management System. SRS refers to Software Requirements Specification. RTM refers to Requirements Traceability Matrix.

2. Test Items

- React front-end (Login page, Student dashboard, Organizer dashboard)
- Node.js / Express.js backend REST API
- MongoDB database (users, events, and registrations collections, accessed via Mongoose)
- Authentication and Role Management module
- Event Creation and Management module
- Event Discovery and Search module
- Event Registration module (including capacity enforcement)
- Registration History module
- Participation Management module
- Attendance module
- Reporting module

3. Features to be Tested

Features mapped to SRS requirement IDs:

AUTH-01: Registered user sign-in

AUTH-05: Role-based access control for students and organizers

AUTH-06: Prevent students from accessing organizer-only functionality

AUTH-07: Reject invalid login credentials

EVENT-01: Organizer creates an event

EVENT-02: Validate mandatory event information

EVENT-05: Organizer cancels an event they manage

EVENT-07: Prevent unauthorized event modification

EVENT-08: Reject invalid event dates and capacity values

DISC-01: Browse available events

DISC-02: Search for available events

REG-01: Authenticated student registers for an event

REG-03: Prevent duplicate registration

REG-04: Prevent registration after the registration deadline

REG-05: Prevent registration when event capacity is reached

HIST-01: Student views registration history

PART-01: Organizer views registered participants

ATT-01: Organizer marks attendance

ATT-03: Store attendance information

STATUS-02: Detect when event capacity is reached

STATUS-03: Prevent registration for full events

REP-01: Generate registration statistics

REP-03: Display total registration information

NFR – Performance: API response time $\leq$ 2 seconds under expected load

NFR – Security: Password protection

4. Features Not to be Tested

- Online payment collection for paid events
- Live integration with PESU Academy.
- Automated certificate generation
- Internal correctness of third-party libraries- assumed tested upstream
- MongoDB Atlas internals (vendor responsibility)

5. Test Approach / Strategy

Levels:

- Unit tests (module-level : Express route handlers, Mongoose schema validation)
- Integration tests (React frontend - Express REST API; Express - MongoDB via Mongoose)
- System tests (end-to-end flows across modules, e.g. student registers -organizer views participant - organizer marks attendance)
- Acceptance tests

Types:

- Functional testing
- Regression testing
- Performance testing
- Usability testing

Entry Criteria: A stable build is available in the shared repository, test data is loaded into the test MongoDB instance, and the test environment is reachable.

Exit Criteria: All planned test cases for the selected features have been executed, no critical or high-severity defects remain open, and the acceptance criteria for each tested requirement are satisfied.

5.1 Security Validation

- Verify passwords are hashed and never stored or logged in plain text

- Verify authenticated sessions/tokens are required on protected routes and rejected when missing, expired, or tampered with.
- Verify HTTPS is enforced in the deployed environment

6. Test Environment

- Hardware: Standard development laptops or a VM. No specialized hardware required.
- Software: PESU-EMS front end (React) and backend (Node.js/Express) builds; MongoDB test/staging instance.
- Tools: Jest and React Testing Library (unit tests), Cypress or Selenium (end-to-end UI tests), Postman/Newman (API tests), a lightweight load tool (performance), GitHub Issues (defect tracking).
- Test Data: Seeded dummy student and organizer accounts, sample events at varying capacities and statuses (upcoming, full, cancelled), and sample registrations/attendance records.

7. Test Schedule

Milestones:

- Test case design: 10-Oct-2026
- Environment setup: 25-Oct-2026
- Test execution start: 25-Oct-2026
- Test execution end: 20-Nov-2026
- UAT: 10-Oct-2026 to 20-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual and automated) mapped in the RTM
- Test scripts (Jest, Cypress/Selenium, Postman collections)
- Seeded test data set
- Test execution logs
- Defect reports (GitHub Issues)
- Test summary report

9. Roles and Responsibilities

Role	Name	Responsibility
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QA Lead	Shayna Zaidi Gupta	Prepare the test plan, coordinate execution
Test Engineer	Sham Santosh: Registration & Participation Shaurya Ishan: Authentication Shriya C Rao: Attendance & Reporting Shayna Zaidi Gupta: Event Management	Design & execute test cases, log defects
Developer	Shayna Zaidi Gupta Shaurya Ishan Sham Santosh Shriya C Rao	Fix defects logged against their own module and support triage
Product Owner / Reviewer	Sham Santosh	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in authentication build	Build and test login first and share early builds with the team
Students accessing organizer-only features	Test role checks on both the UI and the API
Secrets (passwords) leaked in code	Store them in a .env file that is never committed

11. Assumptions & Dependencies

- The MongoDB instance will be stable and reachable throughout the test window.
- Test data will be prepared before execution begins.
- All four team members will be available to test and fix defects in their own module within our schedule.
- Test execution will use the same college-credential authentication flow.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for >4 hours
- Build is too unstable (blocks >30% test cases)

Resume testing if:

- Blocking defects are resolved
- Environment stabilized

13. Test Case Management & Traceability

This includes mapping some of our SRS requirements to test cases.

<u>Req. ID</u>	<u>Feature</u>	<u>Test Case ID</u>
AUTH-01	Registered user sign-in	TC-AUTH-01
AUTH-06	Prevent students from accessing organizer-only functionality	TC-AUTH-06
AUTH-07	Reject invalid credentials	TC-AUTH-07
EVENT-01	Organizer creates an event	TC-EVENT-01
EVENT-02	Require mandatory event information	TC-EVENT-02
EVENT-05	Organizer cancels an event they manage	TC-EVENT-05
EVENT-07	Prevent unauthorized event modification	TC-EVENT-07
EVENT-08	Reject invalid event dates or capacity values	TC-EVENT-08
DISC-01 /DISC-02	Browse and search available events	TC-DISC-01,TC-DISC-02
REG-01	Authenticated student registers for an event	TC-REG-01
REG-03	Prevent duplicate registration	TC-REG-03
REG-04	Prevent registration after the deadline	TC-REG-04
REG-05	Prevent registration when capacity is reached	TC-REG-05
HIST-01	Student views their registration history	TC-HIST-01
PART-01	Organizer views registered participants	TC-PART-01
ATT-01 / ATT-03	Organizer marks and stores attendance	TC-ATT-01, TC-ATT-03
STATUS-02 / STATUS-03	Capacity detection and full-event registration block	TC-STATUS-02, TC-STATUS-03
REP-01 / REP-03	Registration statistics and totals	TC-REP-01, TC-REP-03
NFR- Performance	API response time <= 2s	TC-PERF-01

NFR-Security	Password hashing	TC-SEC-01
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14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

A brief execution status update after each testing session, and a final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Shayna Zaidi Gupta	16-09-2026
Dev Lead	Shaurya Ishan	16-09-2026
Product Owner	Sham Santosh	16-09-2026